Roll No……………

END TERM EXAMINATION NOV DEC 2025

B. Tech (ME) IIIrd Semester

Microprocessor and Microcontroller TME 311

Time: 3.00 Hrs Maximum Marks: 100

Note: -

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20 marks)	**CO-01, C0-02**
(a)	Explain Harvard and Von-Neumann CPU architectures.	
(b)	Compare RISC and CISC CPU architectures.	
(c)	What are the different types of computer software?	
Q2	(20 marks)	**CO-02-CO-03**
(a)	Describe the architecture of the 8051 microcontroller.	
(b)	Explain different addressing modes used in 8051 microcontrollers.	
(c)	List and describe various 8051 instructions including: a. Data transfer instructions b. Arithmetic instructions	
Q3	(20 marks)	**CO-03, CO-04**
(a)	Provide an overview of Intel 8085 to Intel Pentium processors.	
(b)	Draw and explain the pin diagram of the 8085 microprocessor.	
(c)	Explain the external system bus architecture of 8085 microprocessor.	
Q4	(20 marks)	**CO-04**
(a)	What are the general-purpose registers and flags in 8085 microprocessor?	
(b)	Differentiate between peripheral I/O and memory-mapped I/O.	

(c)	Describe the instruction and data format used in 8085 assembly language,	
Q5	(20 marks)	**CO-05&06**
(a)	Write an assembly language program to add two 8 bit numbers stored at address 2050 and address 2051 in 8085 microprocessor. The starting address of the program is taken as 2000.	
(b)	Write an assembly language program to swap two 8-bit numbers stored in an 8085 microprocessor.	
(c)	Multiply two 8 bit numbers stored at address 2050 and 2051. Result is stored at address 3050 and 3051. Starting address of program is taken as 2000.	